

Year 12

Mathematics Extension 2

HSC Trial Examination

2011

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided on the back page of this question paper
- All necessary working should be shown in every question

Note: Any time you have remaining should be spent revising your answers.

Total marks – 120

- Attempt Questions 1 – 8
- All questions are of equal value
- Start each question in a new writing booklet
- Write your examination number on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

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Total Marks – 120**Attempt Questions 1 - 8****All questions are of equal value**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

Question 1 (15 marks)**Marks**

a) Evaluate:

i) By completing the square, find $\int \frac{2}{x^2 + 4x + 13} dx$ **2**

ii) Use integration by parts to evaluate $\int 3xe^x dx$. **2**

iii) Evaluate $\int_0^1 xe^{-x^2} dx$ **2**

b) Use the substitution $t = \tan \frac{\theta}{2}$ to evaluate **4**

$$\int_0^{\frac{\pi}{2}} \frac{1}{\cos \theta + 2 \sin \theta + 3} d\theta$$

Answer correct to 3 significant figures.

c) i) Find real numbers a , b and c such that:

$$\frac{7x+4}{(x^2+1)(x+2)} = \frac{ax+b}{x^2+1} + \frac{c}{x+2} \quad \mathbf{3}$$

ii) Hence find $\int \frac{7x+4}{(x^2+1)(x+2)} dx$ **2**

End of Question 1.

Question 2 (15 marks) Use a separate writing booklet**Marks**

- a) Given $A = 3 - 4i$ and $B = 5 + 3i$, express the following in the form $x + iy$, where x and y are real numbers.
- i) $B - A$ **1**
- ii) $\overline{AB}$ **2**
- iii) $\frac{A}{B}$ **2**
- iv) $\sqrt{A}$ **2**
- b) If $z = 1 - \sqrt{3}i$,
- i) Express z in mod-arg form. **1**
- ii) Show that z^6 is an integer. **2**
- c) On the Argand diagram, sketch the region where the inequalities $2 \leq |z| \leq 5$ and $\arg \frac{\pi}{6} < \arg \frac{2\pi}{3}$ hold simultaneously. **3**
- d) **2**
- The points A and B are drawn on an Argand Diagram and are represented by the lines p and q respectively.
- Copy this diagram into your answer booklet.
- On this diagram plot the points $C(-q)$ and $D(p - q)$.

End of Question 2.

Question 3 (15 marks) Use a separate writing booklet**Marks**

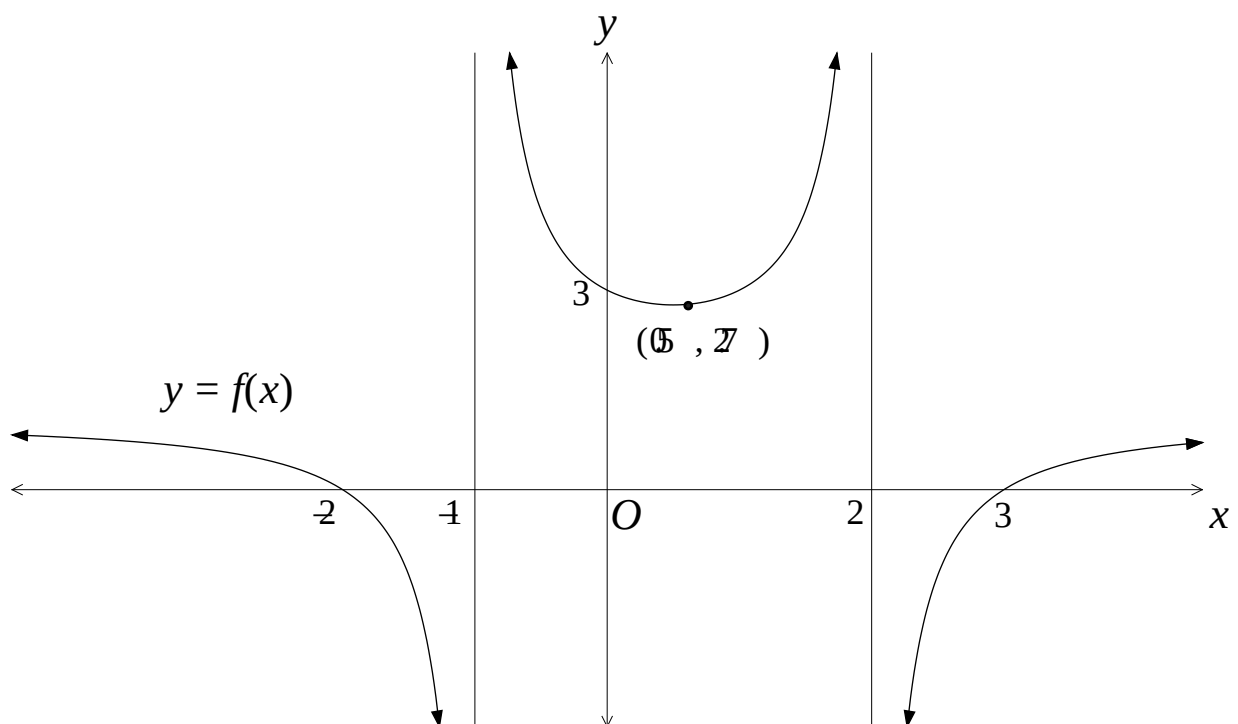
- a) i) Sketch the curve $f(x) = (x+1)(x-2)(x+3)$ showing the intercepts with the coordinate axes. **2**
- ii) On the same diagram, sketch the graph of $y = x - f(x)$ **2**
- iii) The area bounded by $y = f(x)$, the x -axis and the ordinates $x = -3$ and $x = -1$ is rotated about the y -axis. **3**
- Use cylindrical shells to find the volume of the solid of revolution formed.
- b) A particle of unit mass moves in a straight line against a resistance equal to $v + v^2$ where v is its velocity. Initially the particle is at the origin and is travelling with velocity q where $q > 0$.
- i) Show that v is related to displacement by the formula $x = -\ln(1+v) + c$ **2**
- ii) Show that the time t which has elapsed when the particle is travelling with velocity v is given by:

$$t = \ln \frac{q(1+v)}{v(1+q)}$$
 3
- iii) Find v as a function of t . **2**
- iv) Find the value of v as $t \rightarrow \infty$. **1**

End of Question 3.

Question 4 (15 marks) Use a separate writing booklet**Marks**

- a) The equation $x^4 + 2x^3 - 7x^2 - 20x - 12 = 0$ has a double root. Find this root and hence solve this equation. **3**
- b) The equation $x^3 - 4x^2 + 2x - 7 = 0$ has roots α , β and γ . Find an equation which has roots $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$. **2**
- c) The graph of $y = f(x)$ is shown below.



On separate axes draw sketches of the following, showing any critical features.

i) $y = \frac{1}{f(x)}$

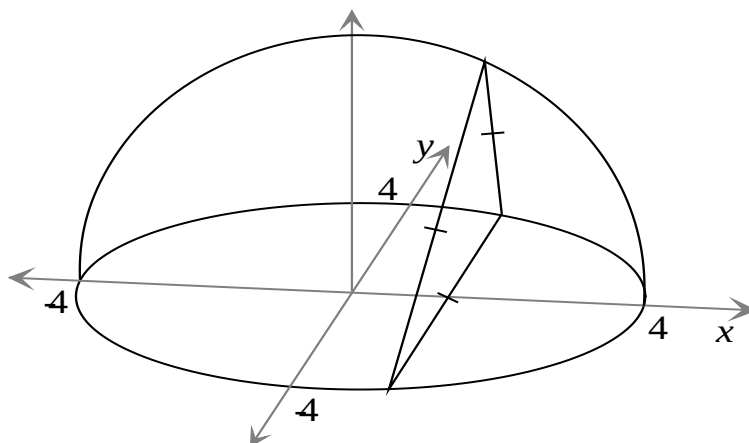
ii) $y = f'(x)$

iii) $y = \pm \sqrt{f(x)}$

Question 4 continues on the next page

Question 4 continued.

d)

4

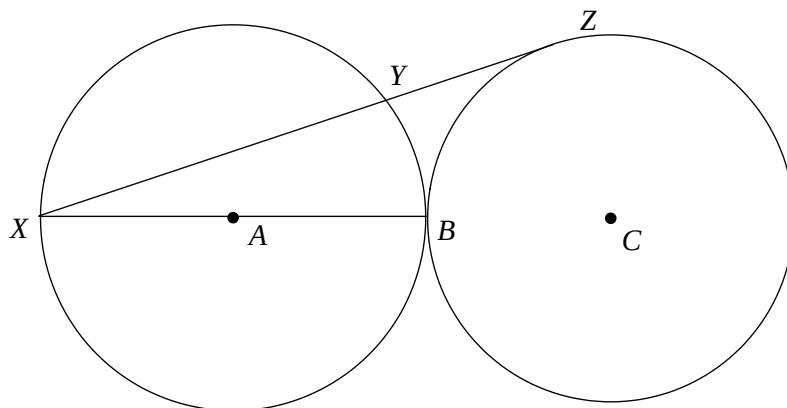
The diagram above shows a solid which has the circle $x^2 + y^2 = 16$ as its base. The cross-section perpendicular to the x axis is an equilateral triangle. Calculate the volume of the solid.

End of Question 4.

Question 5 (15 marks) Use a separate writing booklet**Marks**

- a) The cubic $y = x^3$ is rotated about the y axis $\{x : 0 \leq x \leq 2\}$ to form a solid. Calculate the volume of this solid using the method of slicing. **3**

- b) **3**



Two equal circles touch externally at B . XB is a diameter of one circle. XZ is the tangent from X to the other circle and cuts the first circle at Y . Prove that $2XZ = 3XY$.

- c) Use the principle of Mathematical induction to prove that: **3**

$$\frac{d}{dx} [x^2 + 1]^n = 2xn [x^2 + 1]^{n-1} \text{ for } n \geq 1, n \in \mathbb{Z}.$$

Note: You should not use the function of a function rule (chain rule) as part of your proof.

- d) i) Derive the reduction formula **2**

$$\int x^m (\ln x)^n dx = \frac{x^{m+1} (\ln x)^n}{m+1} - \frac{n}{m+1} \int x^m (\ln x)^{n-1} dx$$

- ii) Hence or otherwise, find $\int_1^e x^3 (\ln x)^3 dx$ **4**

End of Question 5.

Question 6 (15 marks) Use a separate writing booklet**Marks**

- a) A hyperbola has equation $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
- i) Verify that the point $P(a \sec \theta, b \tan \theta)$ lies on the hyperbola. **2**
- ii) The normal to the hyperbola at P cuts the x-axis at M and N is the foot of the perpendicular from P to the x-axis. Prove that the equation of the normal at P is:

$$ax \sin \theta + by = (a^2 + b^2) \tan \theta.$$
 3
- iii) Show that $OM = e^2 ON$, where O is the origin and e is the eccentricity of the hyperbola. **3**
- iv) Prove that $SM = e \times SP$, where S is the focus of the hyperbola. **3**
- b) Find the equation of the tangent to the curve $x^2 y + 2x - 2xy = 0$ at the point (1, 2). **2**
- c) Given that $3+i$ is a root of $P(z) = z^3 + az^2 + bz + 10$, where a and b are real numbers, factorise $P(z)$ over real numbers. **2**

End of Question 6.

Question 7 (15 marks) Use a separate writing booklet**Marks**

- a) A local council consists of 6 independents and 5 others aligned to political parties. A committee of 5 members is to be chosen at random.
- i) How many committees of 5 can be chosen? **1**
- ii) How many of these committees will have a majority of independents? **2**
- b) The equation $x^3 - 3x^2 + ax + 8 = 0$ has roots that are in arithmetic sequence. Find the value of a and hence solve the equation. **4**
- c) i) Differentiate $\sin^{-1} x - \sqrt{1-x^2}$ **2**
- ii) Hence show that $\int_0^a \sqrt{\frac{1+x}{1-x}} dx = \sin^{-1} a + 1 - \sqrt{1-a^2}$ for $0 < a < 1$. **1**
- d) Given that $\sin^{-1} x$, $\cos^{-1} x$ and $\sin^{-1}(1-x)$ are acute:
- i) Show that: **3**
- $$\sin(\sin^{-1} x - \cos^{-1} x) = 2x^2 - 1.$$
- ii) Solve: **2**
- $$\sin^{-1} x - \cos^{-1} x = \sin^{-1}(1-x)$$

End of Question 7

Question 8 (15 marks) Use a separate writing booklet**Marks**

- a) i) Write expressions for $\sin(\alpha - \beta)$ and $\cos(\alpha - \beta)$ and hence show that:

3

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

- ii) Hence write an expression for $\tan\left(\alpha - \frac{\pi}{3}\right)$ in terms of $\tan \alpha$.

1

- b) Given $f(x) = x - \log_e(1 + x^2)$

- i) Show that $f'(x) \geq 0$ for all values of x .

3

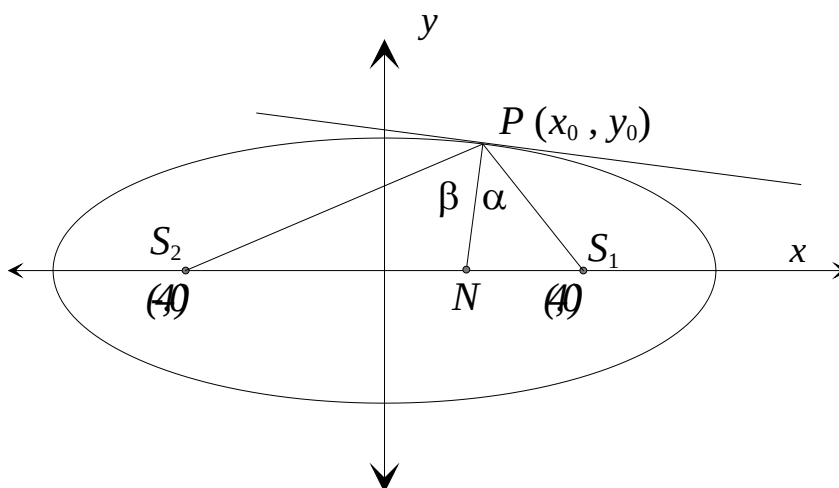
- ii) Hence deduce that $e^x > 1 + x^2$ for all positive values of x .

3

- c) i) Show that the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $P(x_0, y_0)$ has equation: $\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$.

2

- ii)

**3**

In the diagram above, the line PN is the normal to the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ at $P(x_0, y_0)$ and S_1 and S_2 are the foci of the ellipse. $\angle NPS_1 = \alpha$ and $\angle NPS_2 = \beta$. Show that $\alpha = \beta$.

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

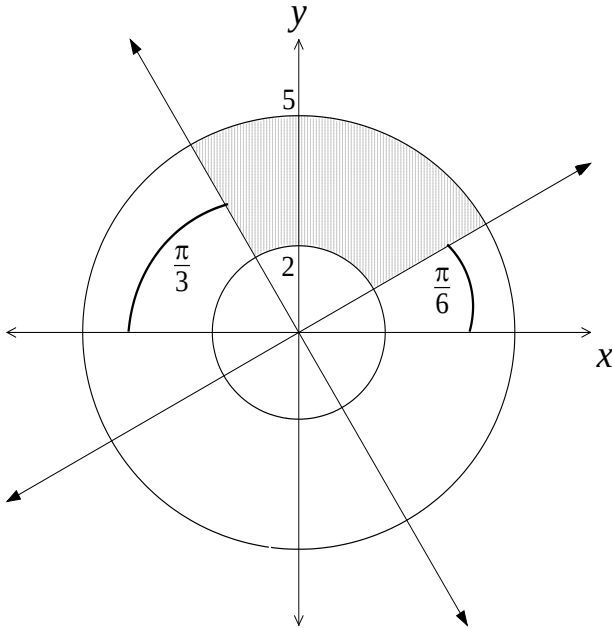
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Question 1	Trial HSC Examination - Mathematics Extension 2	2011
	Solution	Criteria
1(a) (i)	$\int \frac{2}{x^2 + 4x + 13} dx = 2 \int \frac{dx}{(x + 2)^2 + 3^2}$ $= \frac{2}{3} \tan^{-1} \frac{(x + 2)}{3} + c$	2 Marks: Correct answer. 1 Mark: Correctly completes the square
1(a) (ii)	$\int 3xe^x dx = 3 \int x \frac{d}{dx} (e^x) dx$ $= 3(xe^x - \int e^x dx)$ $= 3xe^x - 3e^x + c$	2 Marks: Correct answer. 1 Mark: Set up of the integration by parts.
1(a) iii)	$\int_0^1 xe^{-x^2} dx = -\frac{1}{2} \int_0^1 -2xe^{-x^2} dx$ $= -\frac{1}{2} \left[e^{-x^2} \right]_0^1$ $= -\frac{1}{2} (e^{-1} - e^0)$ $= \frac{1}{2} \left(1 - \frac{1}{e} \right)$ $= \frac{e - 1}{2e}$	2 Marks: Correct answer. 1 Mark: Integrates correctly
1(b)	$t = \tan \frac{\theta}{2}$ $dt = \frac{1}{2} \sec^2 \frac{\theta}{2} d\theta$ $dt = \frac{1}{2} (1 + t^2) d\theta$ $d\theta = \frac{2}{1 + t^2} dt$ When $\theta = 0$ then $t = 0$ and when $\theta = \frac{\pi}{2}$ then $t = 1$ $\cos \theta + 2 \sin \theta + 3 = \frac{1 - t^2 + 2(2t) + 3(1 + t^2)}{1 + t^2}$ $= \frac{2(t^2 + 2t + 2)}{1 + t^2}$ $= \frac{2[1 + (t + 1)^2]}{1 + t^2}$	4 Marks: Correct answer. 3 Marks: Correctly determines the primitive function 2 Marks: Correctly expresses the integral in terms of t 1 Mark: Correctly finds $d\theta$ in terms of dt and

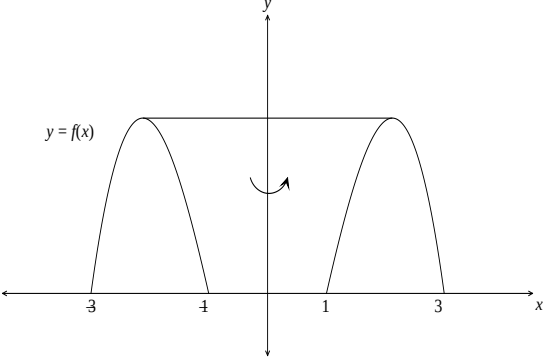
Question 1	Trial HSC Examination - Mathematics Extension 2	2011
	$\int_0^{\frac{\pi}{2}} \frac{1}{\cos \theta + 2 \sin \theta + 3} d\theta = \int_0^1 \frac{1+t^2}{2[1+(t+1)^2]} \times \frac{2}{1+t^2} dt$ $= \int_0^1 \frac{1}{1+(t+1)^2} dt$ $= \left[\tan^{-1}(t+1) \right]_0^1$ $= \tan^{-1} 2 - \frac{\pi}{4}$ $= 0.322$	determines the new limits.
1(c) (i)	$\frac{7x+4}{(x^2+1)(x+2)} = \frac{ax+b}{x^2+1} + \frac{c}{x+2}$ $7x+4 = (ax+b)(x+2) + c(x^2+1)$ Let $x = -2$ and $x = 0$ $-10 = 5c \qquad 4 = b(0+2) - 2(0^2+1)$ $c = -2 \qquad b = 3$ Equating the coefficients of x^2 $0 = a - 2$ $a = 2$ $\therefore a = 2, b = 3$ and $c = -2$	3 Marks: Correct answer. 2 Marks: Calculates two of the variables 1 Mark: Makes some progress in finding a, b or c.
1(c) (ii)	$\int \frac{7x+4}{(x^2+1)(x+2)} dx = \int \left(\frac{2x+3}{(x^2+1)} - \frac{2}{(x+2)} \right) dx$ $= \int \left(\frac{2x}{(x^2+1)} + \frac{3}{(x^2+1)} - \frac{2}{(x+2)} \right) dx$ $= \ln(x^2+1) + 3 \tan^{-1} x - 2 \ln x+2 + c$	2 Marks: Correct answer. 1 Mark: Correctly finds one of the integrals.
	/15	

Question 2		Trial HSC Examination - Mathematics Extension 2		2011
Part	Solution	Marks	Comment	
a)	i. $B - A = (5 + 3i) - (3 - 4i)$ $= 5 + 3i - 3 + 4i$ $= 2 + 7i$ ii. $\overline{AB} = \overline{(3 - 4i)(5 + 3i)}$ $= \overline{15 + 9i - 20i + 12}$ $= \overline{27 - 11i}$ $= 27 + 11i$ iii. $\frac{A}{B} = \frac{3 - 4i}{5 + 3i}$ $= \frac{3 - 4i}{5 + 3i} \times \frac{5 - 3i}{5 - 3i}$ $= \frac{15 - 9i - 20i - 12}{25 + 9}$ $= \frac{3 - 29i}{34} = \frac{3}{34} - \frac{29}{34}i$ iv. Let $\sqrt{A} = x + iy$ (a and b real) $\therefore A = x^2 - y^2 + 2xyi$ $\therefore x^2 - y^2 = 3$ -----(1) $2xy = -4$ $(x^2 + y^2)^2 = (x^2 - y^2)^2 + 4x^2y^2$ $= 3^2 + 4^2$ $= 25$ $x^2 + y^2 = 5$ ----- (2) (1) + (2) $2x^2 = 8 \rightarrow x = \pm 2$ (2) - (1) $2y^2 = 2 \rightarrow y = \pm 1$ Since $2xy = -4$ $\sqrt{A} = \pm(2 - i)$	1 1 1 1 1 1 1	Expansion Conjugate Realising denominator Answer Any fair method Answer	

Question 2		Trial HSC Examination - Mathematics Extension 2		2011
Part	Solution	Marks	Comment	
b)	i. $r = \sqrt{1+3} = 2$ $\tan \theta = \frac{-\sqrt{3}}{1} = -\sqrt{3}$ $\therefore \theta = -\frac{\pi}{3}$ $1 - \sqrt{3}i = 2 \operatorname{cis} \left(-\frac{\pi}{3} \right)$	1	Mod-Arg Form	
	ii. $z = 1 - \sqrt{3}i = 2 \left[\cos \left(-\frac{\pi}{3} \right) + i \sin \left(-\frac{\pi}{3} \right) \right]$ $\therefore z^6 = 64 \left[\cos \left(-\frac{6\pi}{3} \right) + i \sin \left(-\frac{6\pi}{3} \right) \right]$ $= 64 \operatorname{cis} (-2\pi)$ $= 64$	1 1	Use of De Moivre's Solution	
c)		1 1 1	Circles Rays Correct Region	

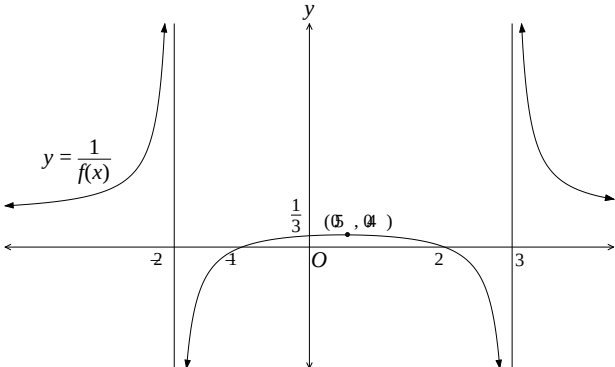
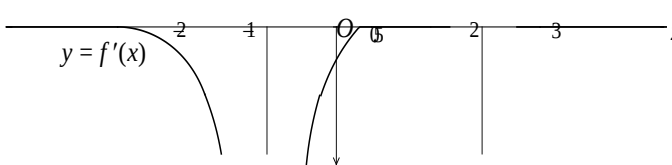
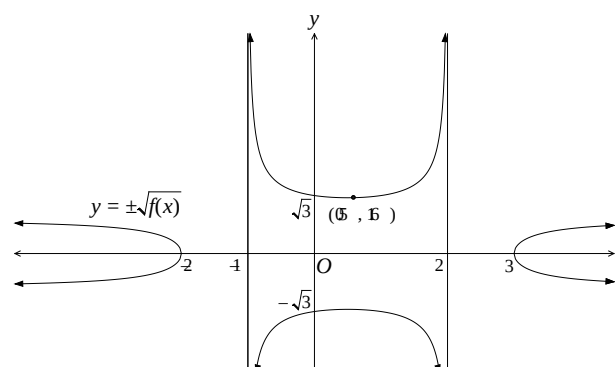
Question 2		Trial HSC Examination - Mathematics Extension 2		2011
Part	Solution	Marks	Comment	
d)	C D	2	1 each for correctly plotting C and D	
		/15		

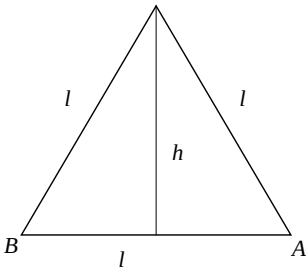
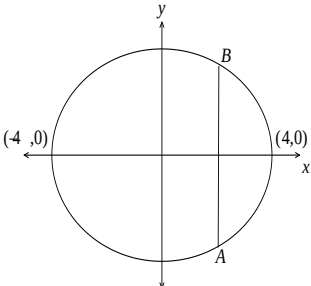
Question 3		Trial HSC Examination - Mathematics Extension 2		2011
Part	Solution	Marks	Comment	
a)	i. $f(x) = (x + 1)(x - 2)(x + 3)$ -6 -6 ii. $y = x - f(x)$	1 1 2	Intercepts Correct shape Deduct a mark for a major feature missing or incorrect,	

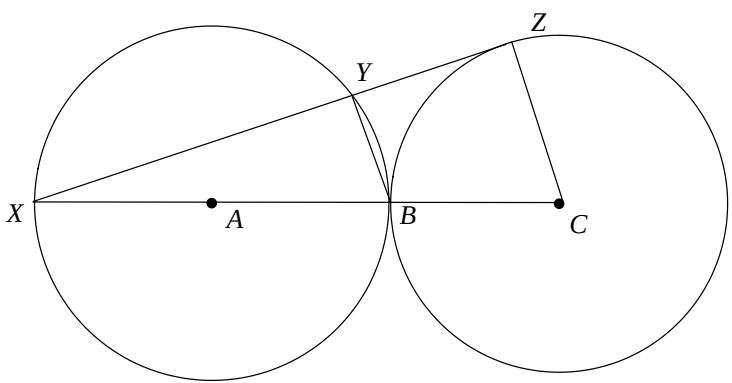
Question	3	Trial HSC Examination - Mathematics Extension 2	2011
Part	Solution	Marks	Comment
	iii.  $f(x) = (x+1)(x-2)(x+3)$ $= (x^3 + 2x^2 - 5x - 6)$ By cylindrical shells $V = \int_a^b 2\pi xy \, dx$ $V = \int_1^3 2\pi x (x^3 + 2x^2 - 5x - 6) \, dx$ $= 2\pi \int_1^3 (x^4 + 2x^3 - 5x^2 - 6x) \, dx$ $= 2\pi \left[\frac{x^5}{5} + \frac{x^4}{2} - \frac{5x^3}{3} - 3x^2 \right]_1^3$ $= 2\pi \left[\left(17\frac{1}{10} \right) - \left(-3\frac{29}{30} \right) \right]$ $= \frac{632\pi}{15}$	1 1 1	
b)	i. $\ddot{x} = -(v + v^2)$ $v \frac{dv}{dx} = -(v + v^2)$ $\frac{dv}{dx} = \frac{-(v + v^2)}{v}$ $\frac{dx}{dv} = \frac{-1}{1+v}$ $x = -\int \frac{1}{1+v} \, dv$ $x = -\ln(1+v) + c$	1 1	

Question	3	Trial HSC Examination - Mathematics Extension 2		2011
Part	Solution		Marks	Comment
	ii. $\frac{dv}{dt} = -(v + v^2)$ $\frac{dt}{dv} = -\frac{1}{(v + v^2)}$ $t = -\int \frac{dv}{(v + v^2)}$ $t = -\int \frac{dv}{v(1 + v)}$ $t = -\int \left(\frac{1}{v} - \frac{1}{1 + v} \right) dv$ $t = \int \left(\frac{1}{(1 + v)} - \frac{1}{v} \right) dv$ $t = \ln(1 + v) - \ln(v) + c$ $t = \ln\left(\frac{1 + v}{v}\right) + c$ When $t = 0, v = q$ $0 = \ln\left(\frac{1 + q}{q}\right) + c$ $c = -\ln\left(\frac{1 + q}{q}\right)$ $c = \ln\left(\frac{q}{1 + q}\right)$ $\therefore t = \ln\left(\frac{1 + v}{v}\right) + \ln\left(\frac{q}{1 + q}\right)$ $t = \ln\left(\frac{q(1 + v)}{v(1 + q)}\right)$			
	$\frac{1}{v(1 + v)} = \frac{A}{v} + \frac{B}{1 + v}$ $1 = A(1 + v) + B(v)$ $v = 0 \quad A = 1$ $1 = 1 + v + Bv$ $Bv = -v \quad \therefore B = -1$			
			1	Expression for t
			1	Value of c
			1	Solution
	iii. Now $e^t = \frac{q(1 + v)}{v(1 + q)}$ $v(1 + q)e^t = q + qv$ $ve^t + qve^t = q + qv$ $ve^t + qve^t - qv = q$ $v(e^t + qe^t - q) = q$ $v = \frac{q}{e^t + qe^t - q}$			
			1	
			1	
			1	Value
	iv. As $t \rightarrow \infty, v \rightarrow 0$			

Question 4		Trial HSC Examination - Mathematics Extension 2		2011
Part	Solution	Marks	Comment	
a)	i. $x^4 + 2x^3 - 7x^2 - 20x - 12 = 0$ If $f(x) = x^4 + 2x^3 - 7x^2 - 20x - 12$ Double root then $f'(x) = 4x^3 + 6x^2 - 14x - 20 = 0$ Test roots of $f'(x)$ Since $f'(-2) = f(-2) = 0$, there is a double root at $x = -2$. Therefore $f(x)$ is divisible by $(x + 2)^2$ i.e. $x^2 + 4x + 4$ By division, $f(x) = (x^2 + 4x + 4)(x^2 - 2x - 3)$ $= (x + 2)^2 (x - 3)(x + 1)$ i.e. Solutions $x = -2, -2, 3, -1$	1 1 1	Double root Division Solution	
b)	$x^3 - 4x^2 + 2x - 7 = 0$ For roots of $\frac{1}{\alpha}, \frac{1}{\beta}$ and $\frac{1}{\gamma}$ $x = \frac{1}{X}$ $\therefore \left(\frac{1}{X}\right)^3 - 4\left(\frac{1}{X}\right)^2 + 2\left(\frac{1}{X}\right) - 7 = 0$ $\frac{1}{X^3} - \frac{4}{X^2} + \frac{2}{X} - 7 = 0$ $1 - 4X + 2X^2 - 7X^3 = 0$ ie. Equation is $7x^3 - 2x^2 + 4x - 1 = 0$	1 1	Substitution Equation	

Question 4	Trial HSC Examination - Mathematics Extension 2		2011
Part	Solution	Marks	Comment
c)	i. $y = \frac{1}{f(x)}$  ii. $y = f'(x)$  iii. $y = \pm \sqrt{f(x)}$ 	2 Each 2 2	Deduct a mark for a major feature missing or incorrect, e.g. asymptotes not correct

Question 4		Trial HSC Examination - Mathematics Extension 2		2011
Part	Solution	Marks	Comment	
d)	 $x^2 + y^2 = 16$$y = \sqrt{16 - x^2}$$\therefore l = 2 \sqrt{16 - x^2}$$\sin 60 = \frac{h}{l}$$h = l \sin 60$$h = \frac{\sqrt{3}}{2} l$$\therefore h = \sqrt{3} \sqrt{16 - x^2}$ $A(x) = \frac{1}{2}bh$$= \frac{1}{2} \left(2\sqrt{16 - x^2} \right) \left(\sqrt{3} \sqrt{16 - x^2} \right)$$= \sqrt{3} (16 - x^2)$ $V = \int_{-4}^4 \sqrt{3} (16 - x^2) dx$$= \sqrt{3} \left[16x - \frac{x^3}{3} \right]_{-4}^4$$= \frac{256\sqrt{3}}{3}$	1 1 1 1	Expression for h Area Integral Answer	
		/15		

Question 5		Trial HSC Examination - Mathematics Extension 2		2011
Part	Solution	Marks	Comment	
5(a)	Area of the slice is a circle radius is x and height δy $A = \pi x^2$ $= \pi (y^{\frac{1}{3}})^2$ $= \pi y^{\frac{2}{3}}$ $\delta V = \delta A \delta y$ $V = \lim_{\delta y \rightarrow 0} \sum_{y=0}^8 \pi y^{\frac{2}{3}} \delta y$ $= \int_0^8 \pi y^{\frac{2}{3}} dy$ $= \pi \left[\frac{3}{5} y^{\frac{5}{3}} \right]_0^8$ $= \frac{3\pi}{5} \times 8^{\frac{5}{3}}$ $= \frac{96\pi}{5} \text{ cubic units}$	3 Marks: Correct answer. 2 Marks: Correct integral for the volume of the solid. 1Mark: Correct expression for the volume of the solid.		
5(b)	 Construction: Join BY, produce XB to C, join CZ. Proof: $\angle XYB = 90^\circ$ (angle in a semicircle is a right angle) $\angle XZC = 90^\circ$ (angle between tangent and radius is a right angle) $BY \parallel CZ$ (corresponding angles are equal) $\triangle XYB \parallel \triangle XZC$ (equiangular) $\frac{XY}{XZ} = \frac{XB}{XC} \text{ (corresponding sides of similar triangles)}$ However $\frac{XB}{XC} = \frac{2}{3}$ ($BC = \frac{1}{2}XB$) $\frac{XY}{XZ} = \frac{2}{3}$ $\therefore 2XZ = 3XY$	3 Marks: Correct answer. 2 Marks: Makes significant progress towards the proof. 1 Mark: States a relevant circle theorem property or equivalent statement.		

Question 5		Trial HSC Examination - Mathematics Extension 2		2011
Part	Solution	Marks	Comment	
c)	Prove $\frac{d}{dx} \left[x^2 + 1 \right]^n = 2xn \left[x^2 + 1 \right]^{n-1}$ Test $n = 1$ $\frac{d}{dx} \left[x^2 + 1 \right]^1 = \frac{d}{dx} \left[x^2 + 1 \right]$ $= 2x$ $= 2x(1) \left[x^2 + 1 \right]^{1-1}$ $\therefore \text{ True for } n = 1$ Assume true for $n = k$ i.e. Assume $\frac{d}{dx} \left[x^2 + 1 \right]^k = 2xk \left[x^2 + 1 \right]^{k-1}$ Consider $n = k + 1$ Want to show $\frac{d}{dx} \left[x^2 + 1 \right]^{k+1} = 2x(k+1) \left[x^2 + 1 \right]^k$ $\text{LHS} = \frac{d}{dx} \left[x^2 + 1 \right]^{k+1} = \frac{d}{dx} \left(x^2 + 1 \right) \left[x^2 + 1 \right]^k$ $= \left(x^2 + 1 \right) . 2kx \left(x^2 + 1 \right)^{k-1} + \left(x^2 + 1 \right)^k . 2x$ $= 2kx \left(x^2 + 1 \right)^k + 2x \left(x^2 + 1 \right)^k$ $= 2x(k+1) \left(x^2 + 1 \right)^k$ $\therefore \text{ True for } n = k + 1 \text{ if true for } n = k$ But true for $n = 1$ $\therefore \text{ true for } n = 1 + 1 = 2 \text{ etc}$ Hence by Mathematical induction, true for all $n \geq 1$	1 1 1		
d)	i. $\int x^m (\ln x)^n dx \qquad u = (\ln x)^n \qquad v' = x^m$ $u' = \frac{n(\ln x)^{n-1}}{x} \qquad v = \frac{x^{m+1}}{m+1}$ $= uv - \int vu' dx$ $= \frac{(\ln x)^n x^{m+1}}{m+1} - \int \frac{x^{m+1} n (\ln x)^{n-1}}{x (m+1)} dx$ $= \frac{(\ln x)^n x^{m+1}}{m+1} - \frac{n}{m+1} \int \left[x^m (\ln x)^{n-1} \right] dx$	1 1		

Question 5		Trial HSC Examination - Mathematics Extension 2		2011
Part	Solution	Marks	Comment	
d)	ii. $\int_1^e x^3 (\ln x)^3 dx = \left[\frac{x^4 (\ln x)^3}{4} \right]_1^e - \frac{3}{4} \left(\int_1^e x^3 (\ln x)^2 dx \right)$ $= \left[\frac{x^4 (\ln x)^3}{4} \right]_1^e - \frac{3}{4} \left\{ \left[\frac{x^4 (\ln x)^2}{4} \right]_1^e - \frac{2}{4} \int_1^e x^3 (\ln x)^1 dx \right\}$ $= \left[\frac{x^4 (\ln x)^3}{4} - \frac{3x^4 (\ln x)^2}{16} \right]_1^e + \frac{3}{8} \left\{ \int_1^e x^3 (\ln x)^1 dx \right\}$ $= \left[\frac{x^4 (\ln x)^3}{4} - \frac{3x^4 (\ln x)^2}{16} \right]_1^e + \frac{3}{8} \left\{ \left[\frac{x^4 (\ln x)^1}{4} \right]_1^e - \frac{1}{4} \int_1^e x^3 (\ln x)^0 dx \right\}$ $= \left[\frac{x^4 (\ln x)^3}{4} - \frac{3x^4 (\ln x)^2}{16} + \frac{3x^4 (\ln x)^1}{32} \right]_1^e - \frac{3}{32} \int_1^e x^3 dx$ $= \left[\frac{x^4 (\ln x)^3}{4} - \frac{3x^4 (\ln x)^2}{16} + \frac{3x^4 (\ln x)^1}{32} - \frac{3x^4}{128} \right]_1^e$ $= \left[\frac{e^4}{4} - \frac{3e^4}{16} + \frac{3e^4}{32} - \frac{3e^4}{128} \right] + \left[\frac{3}{128} \right]$ $= \frac{17e^4 + 3}{128}$	1 1 1 1		
		/15		

Question 6		Trial HSC Examination - Mathematics Extension 2		2011
Part	Solution	Marks	Comment	
a)	i. $\frac{(a \sec \theta)^2}{a^2} - \frac{(b \tan \theta)^2}{b^2} = 1$ $LHS = \frac{a^2 \sec^2 \theta}{a^2} - \frac{b^2 \tan^2 \theta}{b^2}$ $= \sec^2 \theta - \tan^2 \theta$ $= 1 + \tan^2 \theta - \tan^2 \theta$ $= 1$ $= RHS$ Therefore P lies on the hyperbola ii. $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ $\therefore \frac{2x}{a^2} - \frac{2y}{b^2} \frac{dy}{dx} = 0$ $\therefore \frac{dy}{dx} = \frac{b^2 x}{a^2 y}$ At $(a \sec \theta, b \tan \theta)$ $\frac{dy}{dx} = \frac{b^2 (a \sec \theta)}{a^2 (b \tan \theta)} = \frac{b \sec \theta}{a \tan \theta}$ $\therefore m = \frac{b}{a} \operatorname{cosec} \theta$ Gradient of Normal = $-\frac{a}{b} \sin \theta$ Equation: $y - b \tan \theta = -\frac{a}{b} \sin \theta (x - a \sec \theta)$ $by - b^2 \tan \theta = -ax \sin \theta + a^2 \sec \theta \sin \theta$ $ax \sin \theta + by = b^2 \tan \theta + a^2 \tan \theta$ $ax \sin \theta + by = (a^2 + b^2) \tan \theta \quad \text{-----}(1)$	1 1 1 1 1		

Question 6		Trial HSC Examination - Mathematics Extension 2		2011
Part	Solution	Marks	Comment	
a)	iii. Coordinates of N are $(a \sec \theta, 0)$ For M, sub $y = 0$ into (1) $ax \sin \theta = (a^2 + b^2) \tan \theta$ $\therefore x = \frac{(a^2 + b^2) \tan \theta}{a \sin \theta}$ $\therefore x = \frac{(a^2 + b^2) \sec \theta}{a}$ Coordinates of M are $\left(\frac{(a^2 + b^2) \sec \theta}{a}, 0 \right)$ $\therefore OM = \frac{(a^2 + b^2) \sec \theta}{a}$ Now $e^2 ON = e^2 (a \sec \theta)$ Also for a hyperbola $b^2 = a^2 (e^2 - 1)$ $\frac{b^2}{a^2} + 1 = e^2$ $\frac{a^2 + b^2}{a^2} = e^2$ $\therefore e^2 ON = \left(\frac{a^2 + b^2}{a^2} \right) a \sec \theta$ $= \frac{(a^2 + b^2) \sec \theta}{a}$ $= OM$	1 1 1		

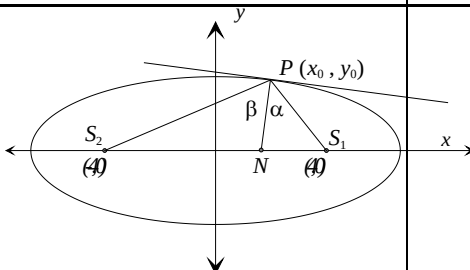
Question 6		Trial HSC Examination - Mathematics Extension 2		2011
Part	Solution	Marks	Comment	
	iii. $SM = ae - \frac{a^2 + b^2}{a} \sec \theta$ $= ae - \frac{a^2 e^2}{a} \sec \theta$ $= ae (1 - e \sec \theta)$ $b^2 = a^2 (e^2 - 1)$ $b^2 = a^2 e^2 - a^2$ $a^2 + b^2 = a^2 e^2$ Now $SP = \sqrt{(ae - a \sec \theta)^2 + (0 - b \tan \theta)^2}$ $= \sqrt{a^2 e^2 - 2ea^2 \sec \theta + a^2 \sec^2 \theta + a^2 (e^2 - 1) \tan^2 \theta}$ $= a \sqrt{e^2 - 2e \sec \theta + \sec^2 \theta + (e^2 - 1)(\sec^2 \theta - 1)}$ $= a \sqrt{e^2 - 2e \sec \theta + \sec^2 \theta + e^2 \sec^2 \theta - e^2 - \sec^2 \theta + 1}$ $= a \sqrt{e^2 \sec^2 \theta - 2e \sec \theta + 1}$ $= a \sqrt{(1 - e \sec \theta)^2}$ $= a (1 - e \sec \theta)$ $\therefore e SP = ae (1 - e \sec \theta)$ $= SM$	1 1 1		
b)	$x^2 y + 2x - 2xy = 0$ $2xy + x^2 \frac{dy}{dx} + 2 - 2y - 2x \frac{dy}{dx} = 0$ $\frac{dy}{dx} (x^2 - 2x) = 2y - 2xy - 2$ $\frac{dy}{dx} = \frac{2y - 2xy - 2}{x^2 - 2x}$ At (1, 2) $\frac{dy}{dx} = \frac{2(2) - 2(1)(2) - 2}{1^2 - 2(1)}$ $= \frac{-2}{-1} = 2$ $y - 2 = 2(x - 1)$ $y - 2 = 2x - 2$ $y = 2x$	1 1	Implicit Differentiation Equation	
c)	All the coefficients of $P(z)$ are real. Then any complex roots occur in conjugate pairs. Since $3 + i$ is a root then $3 - i$ is a root	1	3 – i with correct explanation	

Question 6		Trial HSC Examination - Mathematics Extension 2		2011
Part	Solution		Marks	Comment
	Roots are $3+i$, $3-i$ and α $(3+i)(3-i)\alpha = -\frac{10}{1}$ $(9-i^2)\alpha = -10$ $10\alpha = -10$ $\alpha = -1$ $P(z) = (z- -1)[z-(3+i)][z-(3-i)]$ $= (z+1)(z^2-6z+10)$		1	Correct answer
			/15	

Question 7	Trial HSC Examination - Mathematics Extension 2	2011	
Part	Solution	Marks	Comment
c)i)	$y = \sin^{-1} x - \sqrt{1-x^2}$ $\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}} - \frac{1}{2}(1-x^2)^{-\frac{1}{2}} \times -2x$ $= \frac{1}{\sqrt{1-x^2}} + \frac{x}{\sqrt{1-x^2}}$ $= \frac{1+x}{\sqrt{1-x^2}}$ $= \frac{1+x}{\sqrt{(1+x)(1-x)}}$ $= \frac{\sqrt{1+x}}{\sqrt{1-x}}$ Result defined for $-1 \leq x \leq 1$	1 <	

Question 8		Trial HSC Examination - Mathematics Extension 2		2011
Part	Solution	Marks	Comment	
a)	$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$ $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$ $\tan(\alpha - \beta) = \frac{\sin(\alpha - \beta)}{\cos(\alpha - \beta)}$ $= \frac{\sin \alpha \cos \beta - \cos \alpha \sin \beta}{\cos \alpha \cos \beta + \sin \alpha \sin \beta}$ Divide everything by $\cos \alpha \cos \beta$ $= \frac{\frac{\sin \alpha \cos \beta}{\cos \alpha \cos \beta} - \frac{\cos \alpha \sin \beta}{\cos \alpha \cos \beta}}{\frac{\cos \alpha \cos \beta}{\cos \alpha \cos \beta} + \frac{\sin \alpha \sin \beta}{\cos \alpha \cos \beta}}$ $= \frac{\frac{\sin \alpha}{\cos \alpha} - \frac{\sin \beta}{\cos \beta}}{1 + \frac{\sin \alpha}{\cos \alpha} \frac{\sin \beta}{\cos \beta}}$ $= \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$ ii. $\tan\left(\alpha - \frac{\pi}{3}\right) = \frac{\tan \alpha - \tan \frac{\pi}{3}}{1 + \tan \alpha \tan \frac{\pi}{3}} = \frac{\tan \alpha - \sqrt{3}}{1 + \sqrt{3} \tan \alpha}$	1 1 1 1	1 for the two expressions. Any fair proof Final result	
b)i)	$f(x) = x - \log_e(1+x^2) \quad , \quad 1+x^2 > 0$ $1) \quad f'(x) = 1 - \frac{2x}{1+x^2}$ $= \frac{1+x^2-2x}{1+x^2}$ $= \frac{(x-1)^2}{1+x^2}$ since $(x-1)^2 \geq 0 \quad \forall x$ and $1+x^2 > 0 \quad \forall x$ $f'(x) \geq 0 \quad \forall x$.	3	1 mark correct differentiation. 2 marks correct explanation; 1 mark reasonable attempt.	
ii)	ii) deduce $e^x > 1+x^2$ $f'(x) > 0$ means $f(x)$ monotonic increasing - for $x=0$, $f(0) = 0$ so for $x \geq 0$, $f(x) \geq 0$. so for $x > 0$, $x - \log_e(1+x^2) > 0$ i.e. $x > \log_e(1+x^2)$ i.e. $e^x > 1+x^2, x > 0$.	3	1 mark. 1 mark. 1 mark.	

Question 8		Trial HSC Examination - Mathematics Extension 2		2011
Part	Solution	Marks	Comment	
c)	i. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$ $\frac{2y}{b^2} \frac{dy}{dx} = -\frac{2x}{a^2}$ $\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$ At (x_0, y_0) Equation $y - y_1 = m(x - x_1)$ $m = -\frac{b^2 x_0}{a^2 y_0} \qquad y - y_0 = -\frac{b^2 x_0}{a^2 y_0} (x - x_0)$ $a^2 y y_0 - a^2 y_0^2 = -b^2 x_0^2 + b^2 x x_0$ $b^2 x x_0 + a^2 y y_0 = a^2 y_0^2 + b^2 x^2$ Divide every thing by $a^2 b^2$ $\frac{x x_0}{a^2} + \frac{y y_0}{b^2} = \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2}$ But $\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} = 1$ $\therefore \frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$	1 1	Gradient	

Question 8		Trial HSC Examination - Mathematics Extension 2		2011
Part	Solution	Marks	Comment	
ii)	Ellipse is $\frac{x^2}{25} + \frac{y^2}{9} = 1$ or $9x^2 + 25y^2 = 225$ Equation of tangent is $\frac{xx_0}{25} + \frac{yy_0}{9} = 1$ Differentiate implicitly. $\frac{x_0}{25} + \frac{dy}{dx} \frac{y_0}{9} = 0$ $\frac{dy}{dx} \frac{y_0}{9} = -\frac{x_0}{25}$ $\frac{dy}{dx} = -\frac{9x_0}{25y_0}$ Gradient of normal = $\frac{25y_0}{9x_0}$ $\tan \alpha = \frac{\left \frac{25y_0}{9x_0} - \frac{y_0}{x_0 - 4} \right }{1 + \frac{25y_0}{9x_0} \cdot \frac{y_0}{x_0 - 4}} = \frac{\left \frac{25x_0y_0 - 100y_0 - 9x_0y_0}{9x_0(x_0 - 4)} \right }{\frac{9x_0^2 - 36x_0 + 25y_0^2}{9x_0(x_0 - 4)}}$ $= \frac{\left \frac{16x_0y_0 - 100y_0}{9x_0^2 + 25y_0^2 - 36x_0} \right }{\frac{4y_0(4x_0 - 25)}{225 - 36x_0}}$ since P lies on ellipse $9x_0^2 + 25y_0^2 = 225$ $= \frac{\left \frac{4y_0(4x_0 - 25)}{9(25 - 4x_0)} \right }{\left \frac{4y_0}{9} \right }$ $\tan \beta = \frac{\left \frac{25y_0}{9x_0} - \frac{y_0}{x_0 + 4} \right }{1 + \frac{25y_0}{9x_0} \cdot \frac{y_0}{x_0 + 4}} = \frac{\left \frac{25x_0y_0 + 100y_0 - 9x_0y_0}{9x_0(x_0 + 4)} \right }{\frac{9x_0^2 + 36x_0 + 25y_0^2}{9x_0(x_0 + 4)}}$ $= \frac{\left \frac{16x_0y_0 + 100y_0}{9x_0^2 + 25y_0^2 + 36x_0} \right }{\frac{4y_0(4x_0 + 25)}{9(25 + 4x_0)}}$ since P lies on ellipse $9x_0^2 + 25y_0^2 = 225$ $= \left \frac{4y_0}{9} \right = \tan \alpha$ $\therefore \alpha = \beta$	 Gradient of $PS_1 = \frac{y_0}{x_0 - 4}$ Gradient of $PS_2 = \frac{y_0}{x_0 + 4}$ 1 1 1	1 for individual gradients 1 for expressions for one angle or tan of angle 1 for second angle and equality	
		/15		